

# Assignment 2 (Final Project)

## 1. Requirements

The assignment requires the development of an AI-driven AutoTestDesign tool capable of performing requirements analysis, risk assessment, and systematic test case generation, among other functions, aligning with industry standards like ISTQB Foundation Level principles and the detailed test techniques defined in ISO/IEC/IEEE standards.

### 1.1 AutoTestDesign Requirements

The AutoTestDesign app comprise of functional requirements (1.1.1) and non-functional requirements (1.1.2).

#### 1.1.1 Functional Requirements (FR)

The system may include the following core features:

| ID     | Feature Category               | Description                                                                                                                                                                                                  |
|--------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR 1.0 | Input/Parsing                  | The system can <b>ingest software requirements</b> from various sources (e.g., CSV, plain text, or direct user input).                                                                                       |
| FR 1.1 | Requirement Structuring        | The system can parse and tokenize raw text, identifying key components like <b>Input Fields, Data Ranges, Conditions, and Expected Actions</b> .                                                             |
| FR 2.0 | Risk Analysis & Prioritization | The system can assign a <b>Risk Score</b> and <b>Test Priority</b> (High, Medium, Low) to each imported requirement.                                                                                         |
| FR 3.0 | Black-Box Test Design          | The system can automatically apply and generate test cases for at least three core Black-Box techniques from ISO 29119-4 (e.g., <b>Equivalence Partitioning, Boundary Value Analysis, Decision Tables</b> ). |

| ID     | Feature Category        | Description                                                                                                                                                                                     |
|--------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR 4.0 | White-Box Test Modeling | The system can model system behavior (e.g., <b>State Transition Diagram</b> ) and generate optimal test sequences for a chosen coverage criterion (e.g., <b>All States</b> ).                   |
| FR 5.0 | Test Oracle Generation  | The system can help <b>synthesize the Expected Result</b> for a given requirement and specific test data.                                                                                       |
| FR 6.0 | Output & Export         | The system can generate test artifacts (Test Cases, Test Suites, Risk Scores) in a structured, <b>standard format</b> (e.g., JSON or Excel/CSV) suitable for import into Test Management Tools. |
| FR 7.0 | Test Suite Optimization | The system include an optimization to prioritize or minimize the generated Test Suite based on risk or coverage efficiency.                                                                     |

### 1.1.2. Non-Functional Requirements (NFR)

The non-functional requirements include but not limited to:

- Performance
- Usability (UX/UI)
- Security
- Maintainability and Technology

### 1.2 Project Artifact

1. **AI-driven AutoTestDesign tool**, including the source code, prompts, setup instructions (README) and a video demonstration (20%)
2. **Risk Analysis Report** for the application under test (10%)
3. **Test Plan**, which must cover at least the following aspects (40%):
  - **Project Scope**: Background and overall objectives of the software project
  - **Test Items**: Major functional and non-functional features; description of the system architecture and main components

- **High-level Test Suite Design:** Based on the market requirements document or prototype and the risk analysis, select testing techniques for each test suite
- **Schedule or Checklist:** Show test levels and objectives
- **Organization Chart:** Explain team members' responsibilities
- **Chosen Testing Framework and Rationale**
- **Cost Estimation**
- 4. **Detailed Test Design and Execution Document** for one major feature/module (30%):
  - **Test Case Design:** Design test cases for one selected major feature/module and explain test coverage. Use multiple black-box testing techniques and also white-box techniques.
  - **Test Tool Implementation:** Choose a testing framework for executing the tests; ideally develop test scripts based on the detailed test cases above.
  - **Test Result Analysis:** Summary based on the test results.

## 2. Assessment Criteria

- Understanding of concepts 10%
- Coherence of design and implementation 20%
- Coverage and effectiveness/usefulness 40%
- In-depth analysis (generalizability demonstration, reasoning, etc) 20%
- Presentation 10%

## 2. Presentation

Each group has **15 minutes** to complete the project presentation, covering all aspects listed above. A Q&A session follows. Reviewers will ask questions based on the submitted documents, presentation content, and the software testing fundamentals that students are expected to have learned.

## 3. Submission

Each group must submit the following materials by email to the TA **one day before the presentation date**:

- a) **Project Artifact:** Contains all required content described above (the cover page must include the team ID, full names of all members, and student IDs).

b) **Final Presentation PPT:** The first slide must include the team ID and full names and student IDs of all members.

The report and PPT must be submitted in **PDF format**, and the test scripts must be submitted as a compressed file.

**Submission and Presentation Schedule:**

- **Submission Deadline:** Week 13, Friday, before 17:00 pm
- **Presentation Dates:** Week 14–16, Tuesday/Thursday, 10:00–11:35